Demystifying AI: From Prediction to Creation

Day 1: The Evolution & Anatomy of Artificial Intelligence

Mastering the Building Blocks: ML, Deep Learning, and Gen AI

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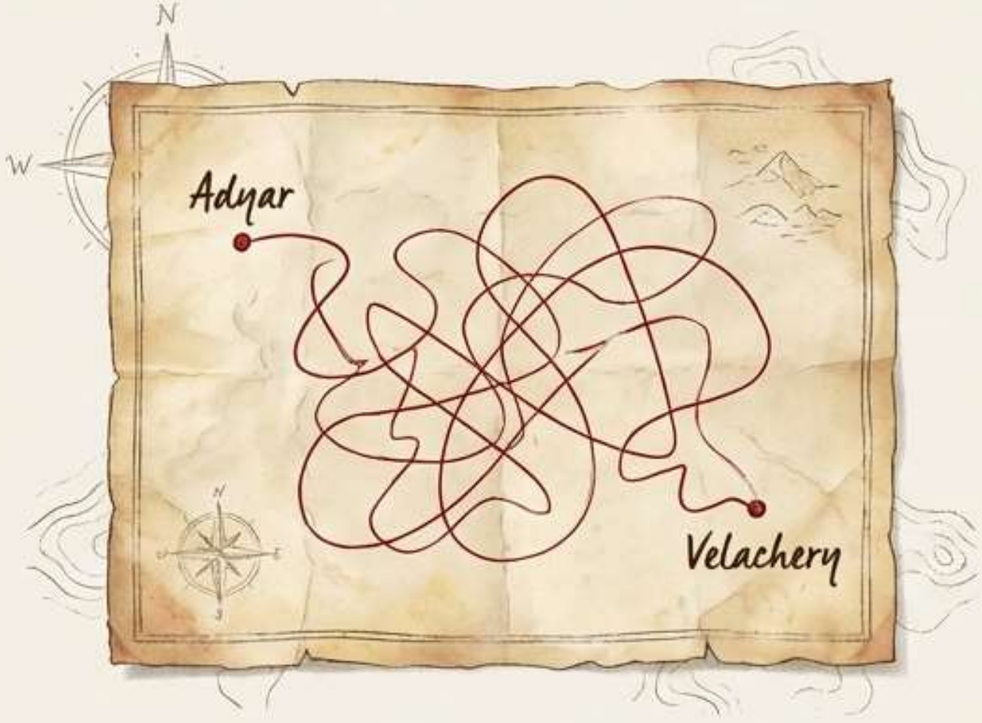
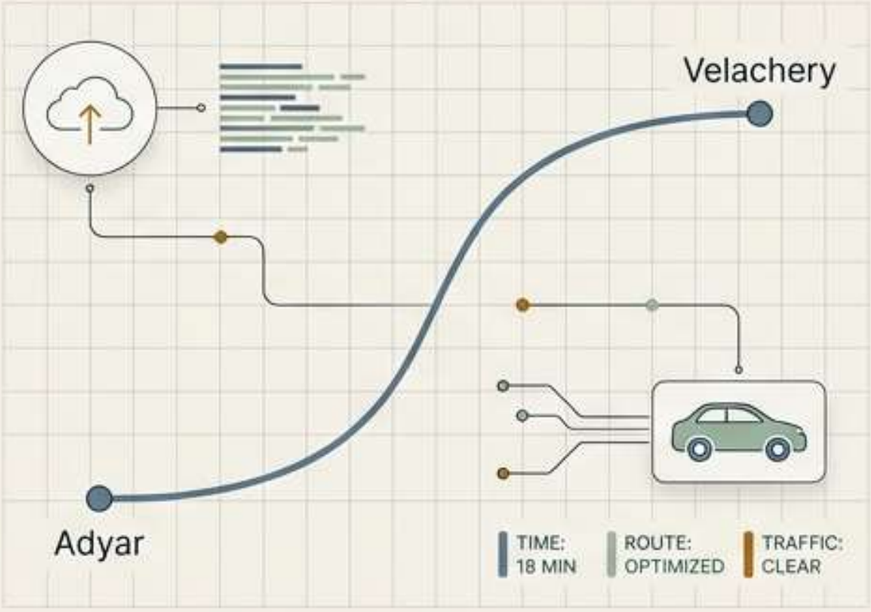
How Does a Machine Mimic Human Behavior?





The Human Limitation

Driving from Adyar to Velachery at 9 AM.

Humans cannot instantly calculate traffic density, shorter routes, and historical patterns with 90-95% accuracy.

The AI Solution

Google Maps processes millions of records, combining current situations with decades of historical data.

The Result: Highly accurate Prediction of time and the shortest route.

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The Architectures of Intelligence

Artificial Intelligence (AI): The overarching concept of machines mimicking human intelligence.

Machine Learning (ML): A subset of AI; uses algorithms to learn from data and make predictions.

Deep Learning (DL): A subset of ML; powered by complex Neural Networks.

Generative AI (Gen AI): A subset of DL; creates entirely new content rather than just predicting

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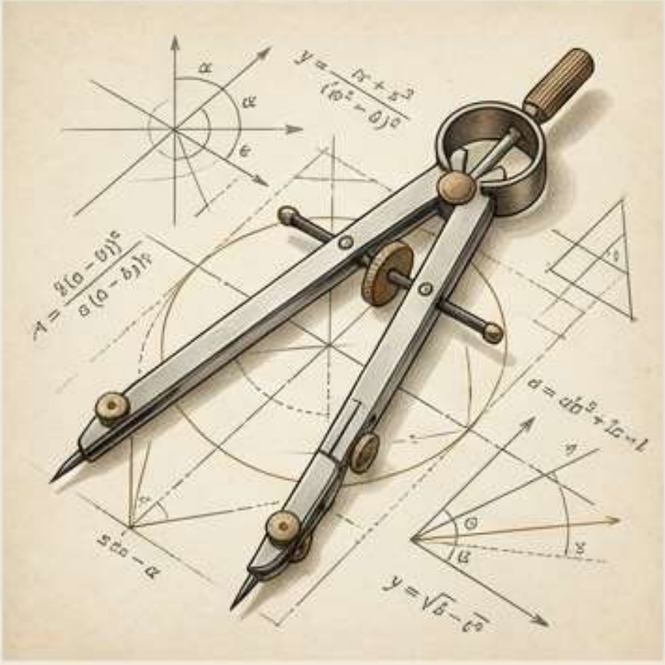
The Core Ingredients of Artificial Intelligence

1. Data (The Fuel)

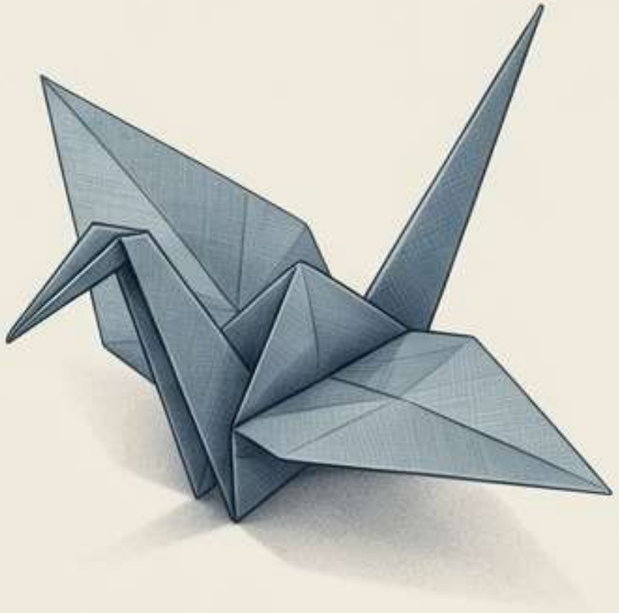
Requires millions to billions of records (e.g., weather data from 1990–2024).

2. Models (The Engine)



The mathematical rules and algorithms that process the data.

3. Output (The Result)



Ranges from predicting numbers to generating art.

Pro Tip: Without massive, high-quality data, even the most advanced models fail. Garbage in, garbage out.

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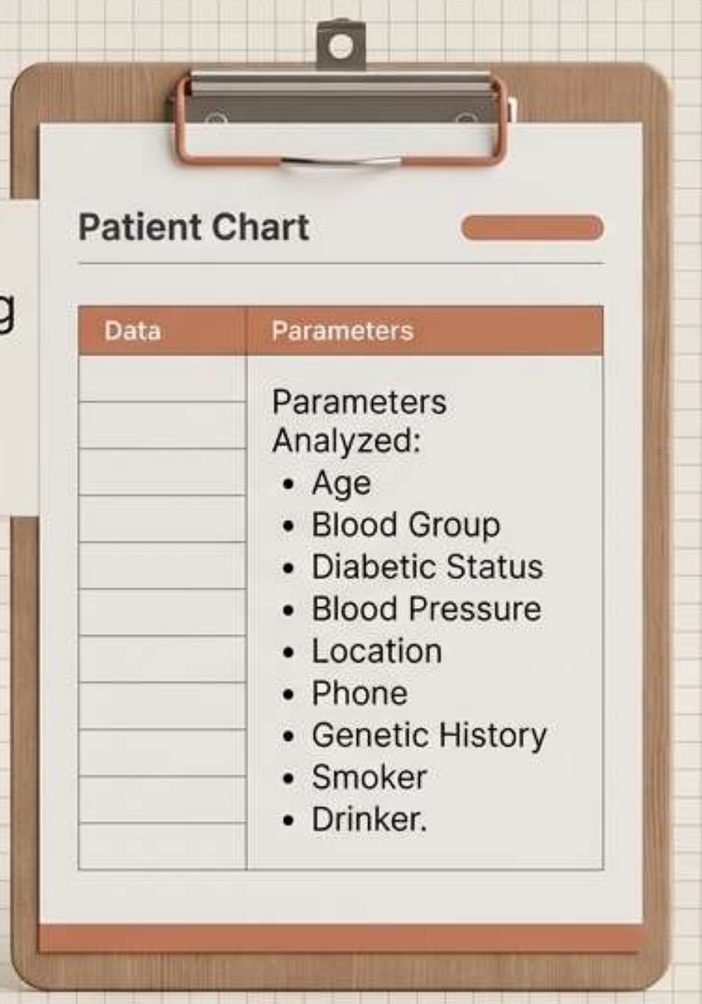
Machine Learning: The Power of Prediction

Use Case:
Predicting Heart Disease Risk



Machine Learning is a subset of AI where machines learn from 1,000s to millions of records.

Core Output: Prediction

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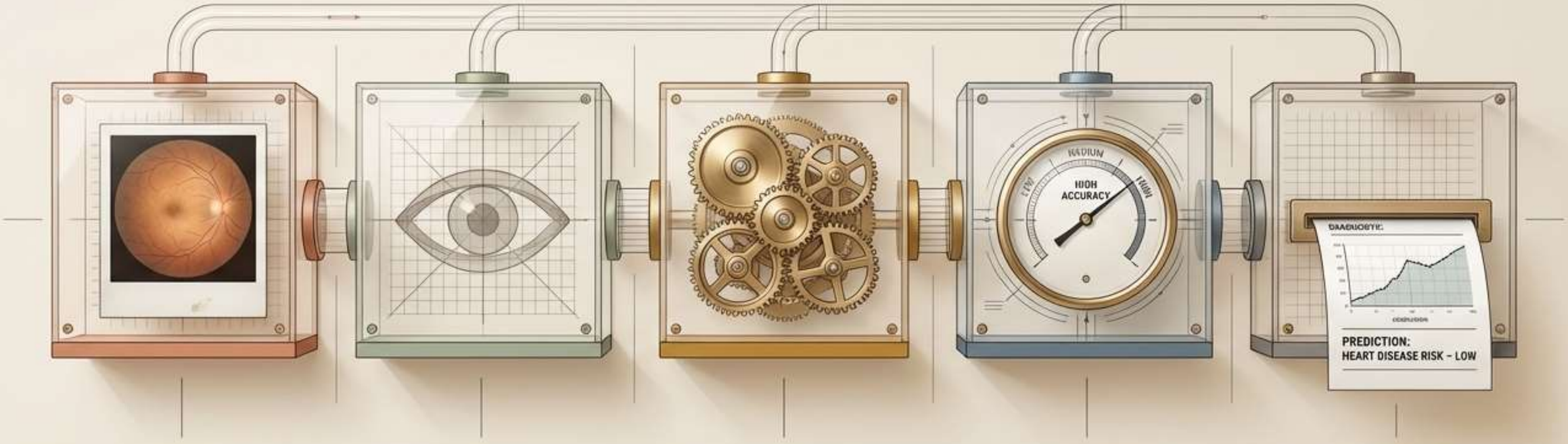
From Raw Data to Accurate Predictions



1. Data Collection

Gathering 1,000s to millions of retinal images.

2. Data Cleansing

Removing errors and standardizing inputs.

3. Model Selection

Choosing the right algorithm (e.g., CNN).

4. Training & Testing

Teaching the model and validating its accuracy.

5. Output

Predicting heart disease risk from a new, unseen image.

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The Golden Rule: Training vs. Testing

80% – Training Data

Used to teach the model the underlying patterns.

20% – Testing & Validation Data

Kept completely hidden during training to test the model's true predictive accuracy.

Did You Know?

The test input must never be part of the training data. Otherwise, the model is just memorizing, not learning!

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ML Types: Supervised Learning (The Guided Path)

Learning from labelled data with clear right/wrong answers.

Path A: Linear Regression

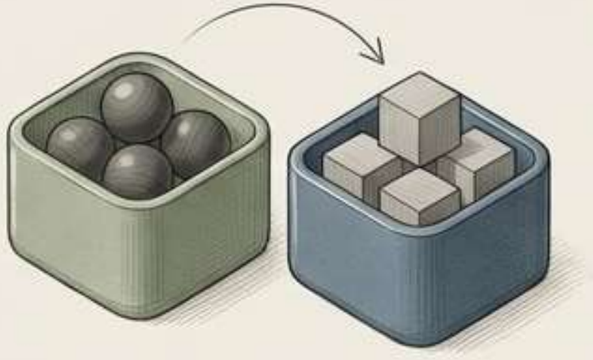
via Terracotta accent

Output: Numerical values.

Examples: Home/Real estate pricing, defect prediction testing.

Path B: Classification

with subtle Slate Blue



Output: Labels or Categories (e.g., Yes or No).

Examples: Google predicting the risk of a heart attack, or defect classification.

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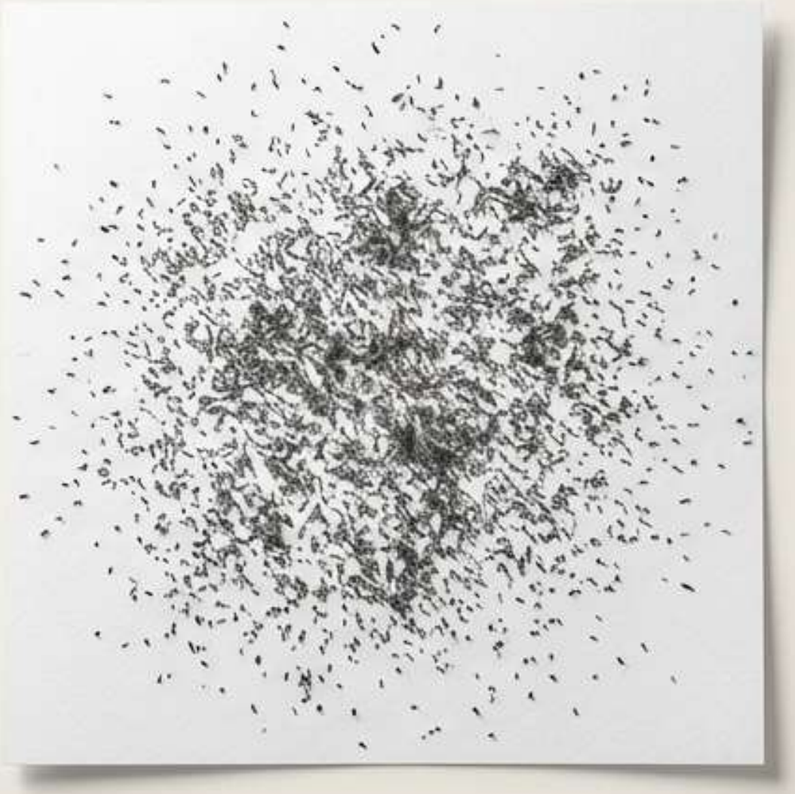
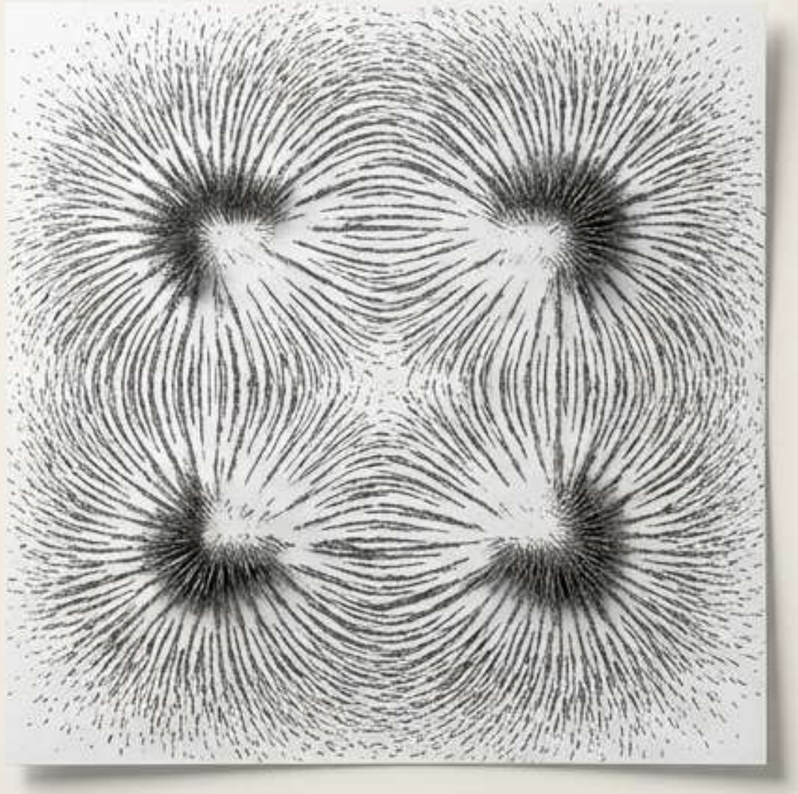
ML Types: Unsupervised Learning (Finding Hidden Patterns)

Unlabelled Data



No explicit feedback, no likes, no comments.

Hidden Structures Discovered



Analyzing behavior to find genres and groups.

Real-World Application: Collaborative Filtering.

How it works: Grouping users and recommending content based entirely on behavioral similarities without explicit labels.

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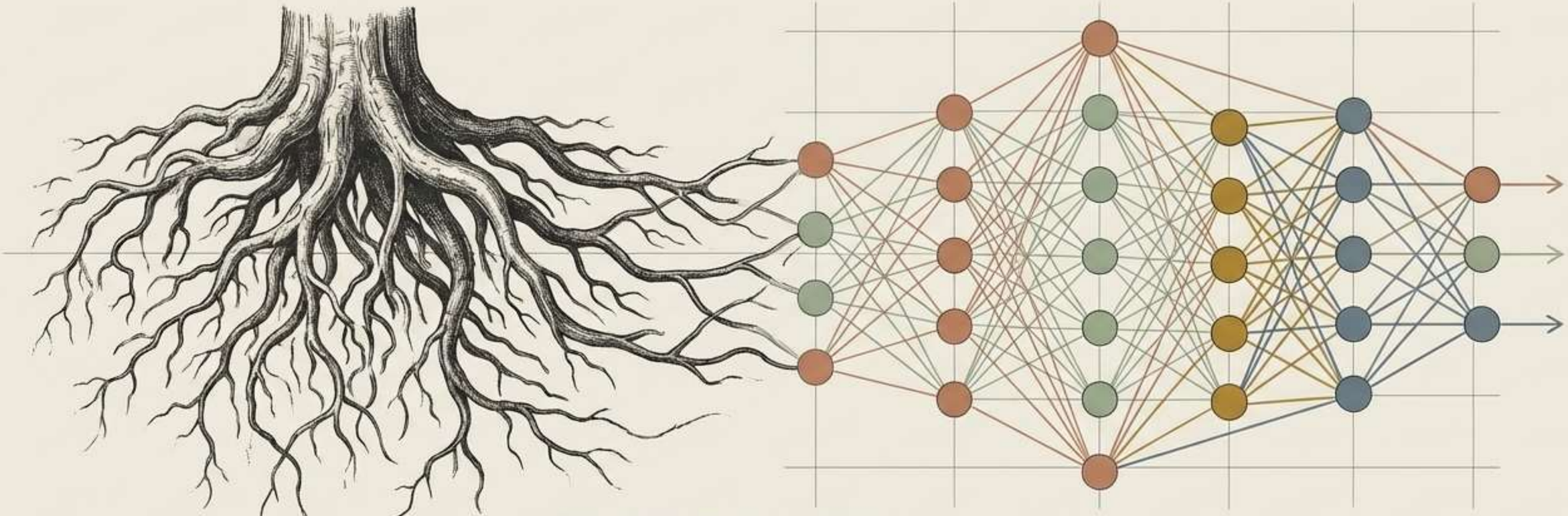
Deep Learning: Mimicking the Human Brain

The Formula: Vast Amounts of Data + Neural Networks = Deep Learning.

Core Function: Still primarily predictive, but capble of handling incredibly complex, multi-layered problems.



Data Scale: Requires millions to trillions of data points to function effectively.

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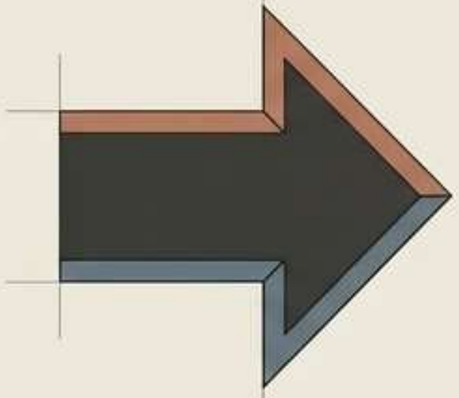
Generative AI: From Prediction to Creation

Predictive Output

Calculating the future based on the past.



Generative Output

Synthesizing entirely new combinations.

The Paradigm Shift: Gen AI does not just retrieve information; it creates Multimodal output.
It can generate totally new Image, Text, Audio, Video, or Code across all domains.

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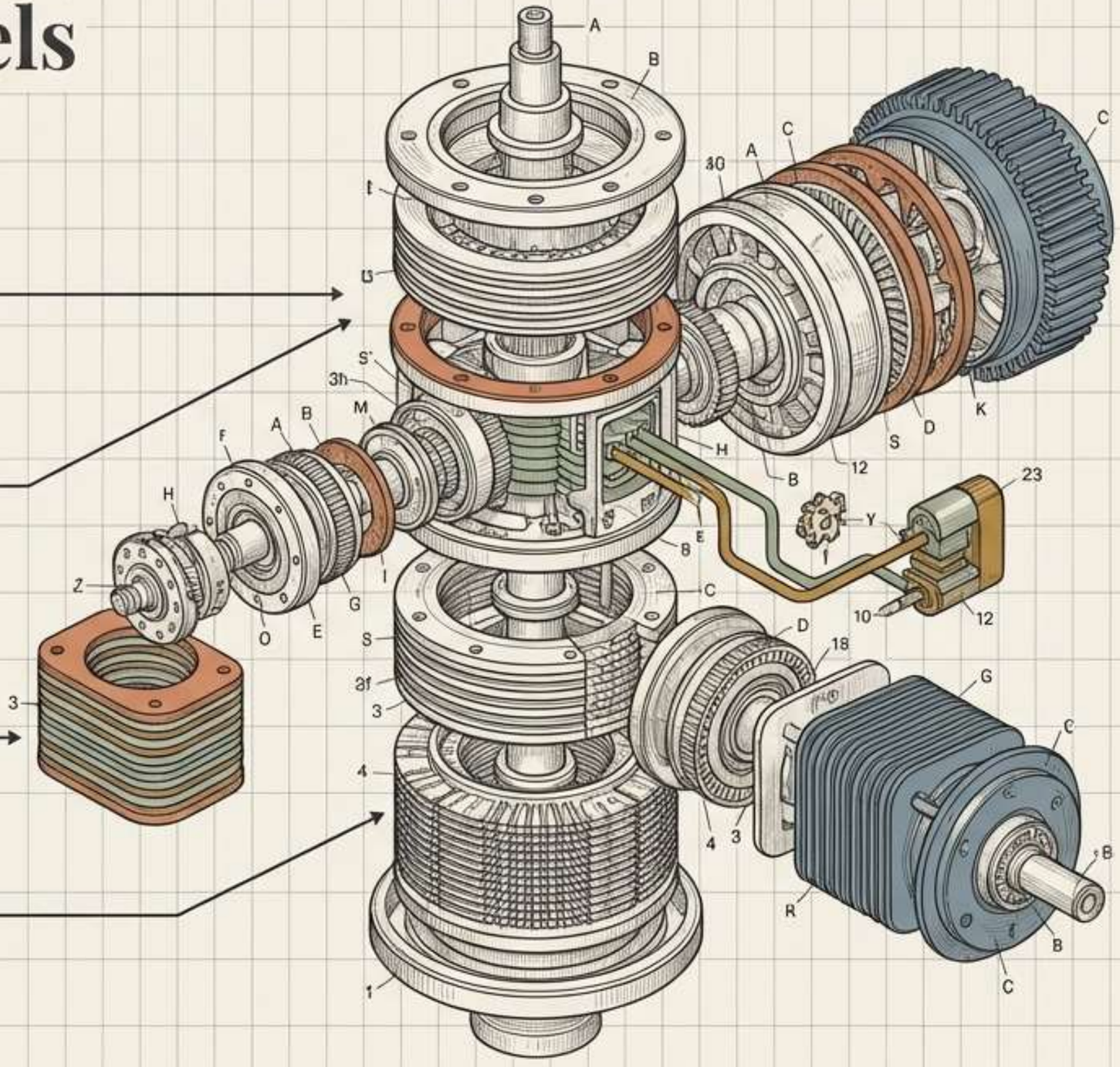
Large Language Models & The GPT Engine

1.
LLM: The engine powering agents like OpenAI's ChatGPT.

2.
GPT: Generative Pre-trained Transformer. Trained on massive, cross-domain datasets.

3.
GPT-4: 96 to 120 Layers.

4.
GPT-5 (Horizon): 190 Layers, 120 Billion Parameters.



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The Artificial Intelligence Landscape Matrix

Artificial Intelligence	Machine Learning	Deep Learning	Generative AI
Broad field of machine intelligence.	Algorithms.	Neural Networks.	LLMs / Transformers.
Variable	Thousands to Millions of records.	Millions to Trillions of records.	Trillions of parameters.
Mimics human behavior.	Predictive.	Predictive (Highly Complex).	Generative (Multimodal).

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Module 1: Knowledge Check

Question 1:
What is the golden split for Training vs. Testing data?

Answer:
80% Training / 20% Testing

Question 2:
Which type of Supervised ML outputs strict numerical values, like real estate prices?

Answer:
Linear Regression

Question 3:
What is the fundamental difference between Deep Learning and Gen AI outputs?

Answer:
Prediction vs. Creation

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End of Day 1. Looking Ahead...

You have successfully mapped the core architectures of the Artificial Intelligence Universe.

Coming up on Day 2:

- Deep Dive: The Anatomy of Large Language Models (LLMs).
- Deconstructing the internal workings of Transformers.

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